ELEC70142

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EXAMINATIONS 2026

Digital VLSI System-on-Chip (SoC) Design

Monday, 11 May 2026, 10:00 AM

Time allowed: 2:00 hours

There are FOUR questions on the paper.

Answer ALL questions.

Any special instructions for invigilators and information for Candidates are on page 1.

Examiners responsible First Marker(s): P.Y.K. Cheung

Second Marker(s): Christos Bouganis

Information for Invigilators:

Students may bring ONE A4 sheet of handwritten notes into the examination.

Information for Candidates:

Students may need red, green, blue, yellow and black coloured pens.

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1. *Figure Q1* shows the layout of a digital circuit using an n-well process. It consists of 14 transistors, labelled in yellow.

   a) What logic functions are performed by transistors T11, T12, T13 and T14?

   [5 marks]

   b) Extract the transistor level schematic circuit for T1 to T4. What logic function does this circuit perform? (Sizes of transistors not required.)

   [5 marks]

   c) Transistors T5 to T10 form a compound gate. Extract the transistor level schematic circuit for T5 to T10. (Sizes of transistors not required.)

   [10 marks]

   d) Based on the answers above, derive the logic equation for OUT3 in terms of input signals A and B.

   [5 marks]

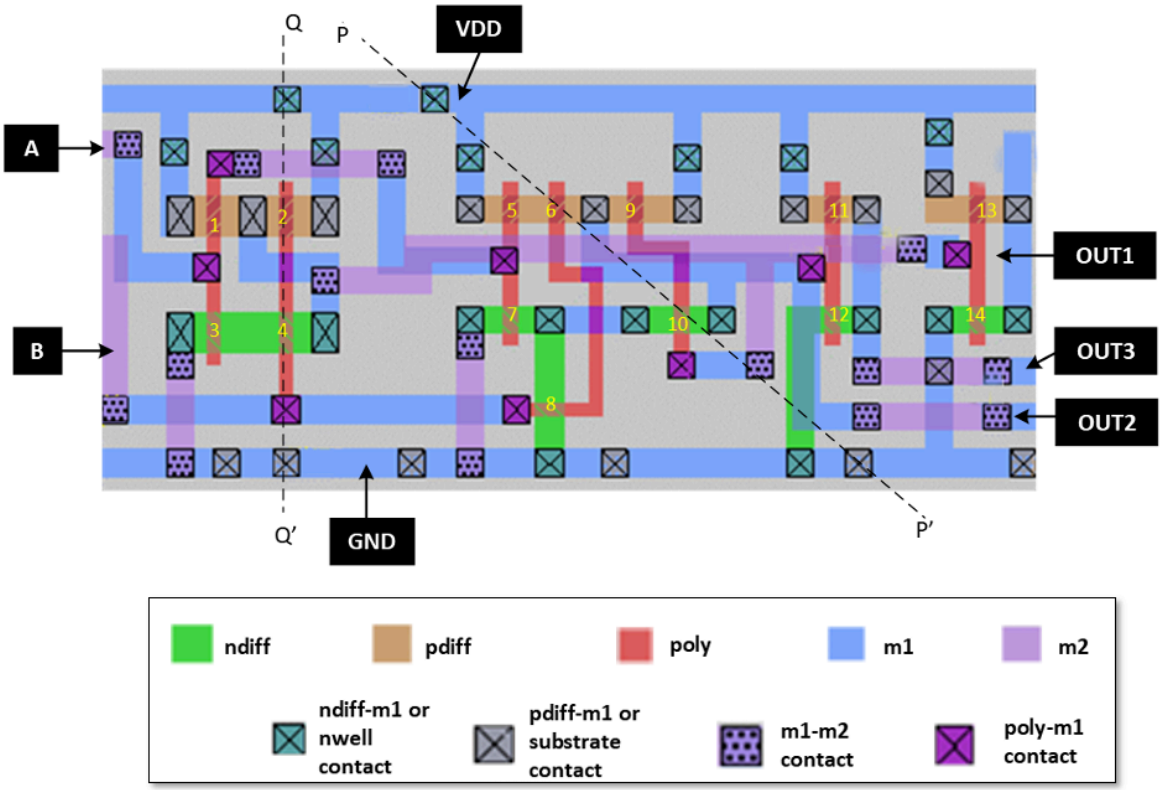


Figure Q1 Layout of a full-custom cell (Q1 & Q2)

2. (a) For the layout shown in *Figure Q1* above, draw the vertical cross section along the line Q and Q′. Label your diagram indicating the n-well region, the different types and levels of doping (e.g. p^-, n^+ etc) and the different metallisation.

   [15 marks]

   (b) Briefly explain the purpose of the nwell and substrate contacts.

   [10 marks]

3. a) Explain briefly the advantages and limitations of the "stuck-at" model used in digital test.

[5 marks]

b) *Figure Q3(a)* shows the schematic circuit of a 2-input gate with a manufacturing defect manifested as a break in the signal track labelled “Open”. Explain why this fault cannot be represented as a stuck-at fault. Derive the faulty behaviour of this circuit as a Boolean equation or a truth-table.

[10 marks]

c) By applying the principle of path sensitisation to the circuit shown in *Figure 3(b)*, generate the test vectors for detecting: i) a stuck-at-0 fault at node F, ii) a stuck-at-1 fault at node G.

[10 marks]

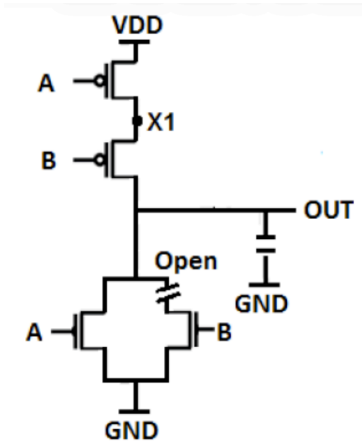


Figure Q3(a)

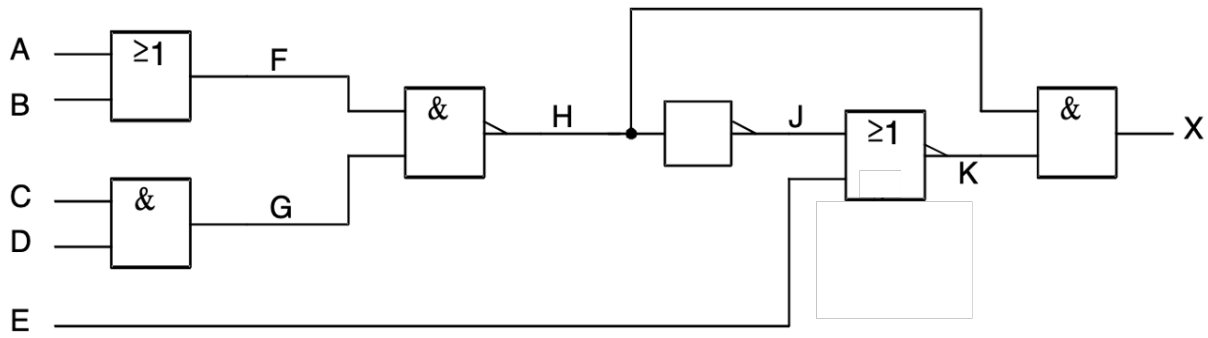


Figure Q3(b)

4. a) Explain the principle behind the operation of a pulse-triggered latch and its advantages and disadvantages.

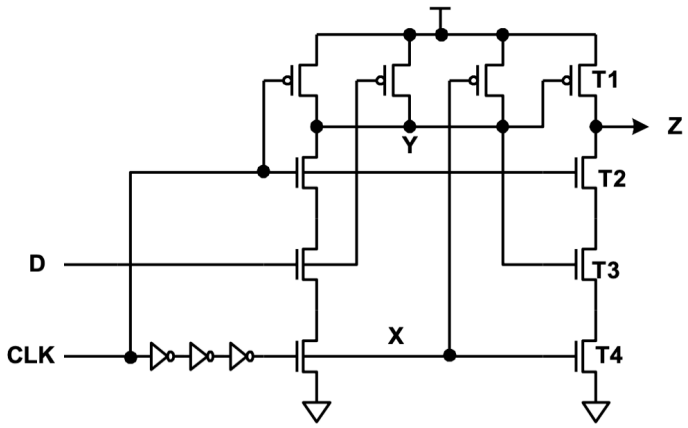
[7 marks]

b) *Figure Q4*(a) shows a synchronous circuit used in the AMD K-6 microprocessor. *Figure Q4(b)* shows the signals at `CLK` and `D`. Draw a timing diagram showing the signals `CLK`, `D`, `X`, `Y` and `Z`, and the state of transistors `T1`, `T2`, `T3` and `T4`. The timing diagram should be labelled with the state of the transistors {on, off} and with the following states for the signals:

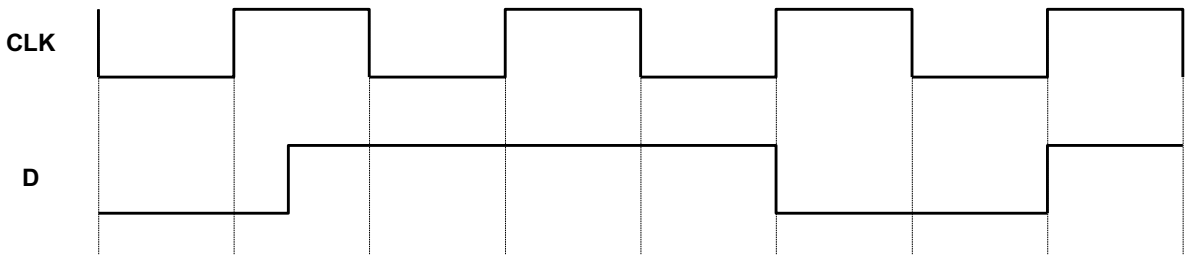
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driven low (DL)	- a path to ground exists,
driven high (DH)	- a path to Vdd exists,
charged low (CL)	- high impedance with no stored charge,
charged high (CH)	- high impedance with stored charge at Vdd,
don't know (X)	- unknown level.

[18 marks]



Question Q4(a)



Question Q4(b)